

## MACKINAC ISLAND, MI, USA

WMO: 727435

Lat: 45.865N

Lon: 84.637W

Elev: 226

StdP: 98.64

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 54820

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -20.1      | -17.2 | -23.6                         | 0.5 | -19.3 | -21.0 | 0.6 | -16.1 | 11.7                  | -7.6 | 10.7 | -7.8 | 3.8                   | 270  | 0.535 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.2                    | 27.2            | 20.5 | 25.8 | 19.6 | 24.0 | 19.0 | 22.3                | 25.0 | 21.3 | 24.0 | 20.3 | 22.9 | 2.4                  | 250  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 21.4                            | 16.5 | 23.8 | 20.3 | 15.4 | 22.8 | 19.1 | 14.3 | 21.8 | 66.7          | 24.9 | 63.1 | 23.7 | 59.4 | 23.1 | 25.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 9.4           | 8.3  | 7.3 | -22.7                      | 29.7 | 4.6                | 1.5 | -26.1                                              | 30.8 | -28.8      | 31.6 | -31.4      | 32.4 | -34.8      | 33.5 |    |    |
| (5)               |      |     | -23.0                      | 23.7 | 4.5                | 1.0 | -26.3                                              | 24.4 | -28.9      | 25.0 | -31.5      | 25.5 | -34.8      | 26.3 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 6.6   | -6.7  | -6.3  | -2.1  | 4.0   | 10.2  | 15.8  | 19.1  | 18.7  | 15.7  | 9.0   | 3.0   | -2.5 | (6)  |
| (7)  |                                                                                  | DBStd       | 10.15 | 5.85  | 5.65  | 5.37  | 4.19  | 3.91  | 3.45  | 2.85  | 2.75  | 3.86  | 4.36  | 4.34  | 4.69 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 2254  | 517   | 457   | 375   | 186   | 45    | 2     | 0     | 0     | 3     | 72    | 212   | 387  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 4413  | 775   | 691   | 632   | 431   | 252   | 90    | 25    | 29    | 96    | 289   | 458   | 645  | (9)  |
| (10) |                                                                                  | CDD10.0     | 1006  | 0     | 0     | 1     | 4     | 52    | 177   | 283   | 269   | 174   | 42    | 3     | 0    | (10) |
| (11) |                                                                                  | CDD18.3     | 125   | 0     | 0     | 0     | 0     | 2     | 15    | 51    | 40    | 17    | 1     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 483   | 0     | 0     | 0     | 0     | 19    | 78    | 199   | 133   | 51    | 3     | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 55    | 0     | 0     | 0     | 0     | 3     | 11    | 25    | 11    | 4     | 0     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 3.1   | 3.7   | 3.6   | 3.2   | 3.3   | 2.9   | 2.5   | 2.4   | 2.3   | 2.6   | 3.3   | 3.6   | 3.8  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 778   | 40    | 29    | 42    | 62    | 76    | 76    | 73    | 76    | 90    | 97    | 68    | 49   | (15) |
| (16) |                                                                                  | PrecMax     | 926   | 84    | 57    | 104   | 150   | 137   | 210   | 128   | 143   | 176   | 207   | 124   | 104  | (16) |
| (17) |                                                                                  | PrecMin     | 580   | 7     | 4     | 2     | 26    | 21    | 20    | 33    | 18    | 37    | 23    | 13    | 5    | (17) |
| (18) |                                                                                  | PrecStd     | 95    | 22    | 14    | 27    | 29    | 28    | 40    | 26    | 35    | 35    | 43    | 32    | 24   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 7.1   | 7.4   | 14.9  | 19.1  | 26.1  | 27.9  | 29.0  | 27.9  | 27.4  | 22.7  | 17.1  | 10.6 | (19) |
| (20) |                                                                                  |             | MCWB  | 4.8   | 5.7   | 10.7  | 12.4  | 17.3  | 20.1  | 21.6  | 21.6  | 21.1  | 18.1  | 12.4  | 9.5  | (20) |
| (21) |                                                                                  | 2%          | DB    | 3.7   | 3.9   | 9.2   | 15.6  | 22.1  | 25.8  | 27.2  | 26.2  | 24.8  | 19.8  | 12.8  | 6.5  | (21) |
| (22) |                                                                                  |             | MCWB  | 2.5   | 2.6   | 6.4   | 9.4   | 15.7  | 19.0  | 20.4  | 20.3  | 19.4  | 15.8  | 10.3  | 4.8  | (22) |
| (23) |                                                                                  | 5%          | DB    | 2.3   | 2.3   | 7.0   | 12.6  | 19.0  | 23.5  | 25.5  | 24.8  | 22.8  | 17.4  | 11.0  | 4.9  | (23) |
| (24) |                                                                                  |             | MCWB  | 1.4   | 1.4   | 4.3   | 7.3   | 13.8  | 17.7  | 19.7  | 19.5  | 18.5  | 14.3  | 9.0   | 3.4  | (24) |
| (25) |                                                                                  | 10%         | DB    | 1.0   | 1.1   | 4.4   | 10.2  | 17.2  | 22.1  | 24.0  | 23.4  | 21.7  | 15.4  | 8.8   | 3.4  | (25) |
| (26) |                                                                                  |             | MCWB  | 0.4   | 0.3   | 2.3   | 5.9   | 12.5  | 16.9  | 19.2  | 19.0  | 18.0  | 12.7  | 7.2   | 2.3  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 5.5   | 5.7   | 11.5  | 14.2  | 19.3  | 22.8  | 23.7  | 23.6  | 22.3  | 20.0  | 13.0  | 9.4  | (27) |
| (28) |                                                                                  |             | MCDB  | 6.6   | 6.9   | 13.4  | 17.3  | 22.6  | 26.1  | 26.6  | 26.5  | 25.0  | 21.7  | 16.1  | 10.1 | (28) |
| (29) |                                                                                  | 2%          | WB    | 2.8   | 2.9   | 7.4   | 10.2  | 16.8  | 20.4  | 22.4  | 21.9  | 20.8  | 16.8  | 10.9  | 5.6  | (29) |
| (30) |                                                                                  |             | MCDB  | 3.7   | 3.6   | 8.9   | 14.2  | 20.4  | 23.7  | 25.0  | 24.4  | 23.3  | 18.2  | 12.4  | 6.5  | (30) |
| (31) |                                                                                  | 5%          | WB    | 1.5   | 1.4   | 4.4   | 8.0   | 15.0  | 19.0  | 21.3  | 20.9  | 19.7  | 14.6  | 8.9   | 3.7  | (31) |
| (32) |                                                                                  |             | MCDB  | 2.1   | 2.0   | 6.3   | 11.2  | 18.1  | 21.8  | 24.0  | 23.5  | 22.0  | 17.0  | 10.3  | 4.7  | (32) |
| (33) |                                                                                  | 10%         | WB    | 0.2   | 0.2   | 2.7   | 6.4   | 13.0  | 17.6  | 20.4  | 19.9  | 18.6  | 13.1  | 7.5   | 2.5  | (33) |
| (34) |                                                                                  |             | MCDB  | 0.8   | 0.9   | 4.4   | 9.6   | 16.3  | 20.6  | 23.0  | 22.6  | 20.8  | 15.1  | 8.5   | 3.4  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 6.1   | 6.8   | 7.4   | 7.9   | 9.1   | 8.8   | 8.2   | 7.8   | 7.6   | 6.4   | 5.3   | 5.1  | (35) |
| (36) |                                                                                  |             | MCDBR | 6.1   | 6.9   | 9.2   | 12.4  | 12.6  | 10.9  | 9.3   | 8.8   | 8.9   | 8.5   | 7.8   | 6.3  | (36) |
| (37) |                                                                                  |             | MCWBR | 5.5   | 5.9   | 6.8   | 7.7   | 7.5   | 6.2   | 5.1   | 5.1   | 5.3   | 6.1   | 6.4   | 5.7  | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 5.8   | 6.4   | 8.8   | 11.3  | 11.1  | 8.9   | 7.8   | 7.4   | 7.9   | 7.9   | 7.3   | 5.8  | (38) |
| (39) |                                                                                  |             | MCWBR | 5.4   | 5.8   | 6.9   | 7.8   | 7.8   | 6.1   | 5.1   | 5.2   | 5.6   | 6.3   | 6.3   | 5.5  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.298 | 0.292 | 0.316 | 0.357 | 0.372 | 0.390 | 0.389 | 0.380 | 0.367 | 0.333 | 0.301 | 0.279 | (40) |      |
| (41) |                                                                                  | taud        | 2.331 | 2.361 | 2.398 | 2.335 | 2.371 | 2.355 | 2.379 | 2.402 | 2.427 | 2.503 | 2.522 | 2.457 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 798   | 890   | 916   | 901   | 895   | 877   | 873   | 868   | 850   | 834   | 805   | 796   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 76    | 90    | 101   | 118   | 118   | 120   | 116   | 109   | 98    | 79    | 64    | 61    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.12  | 2.05  | 3.43  | 4.57  | 5.63  | 6.09  | 6.13  | 5.20  | 3.93  | 2.29  | 1.20  | 0.84  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.12  | 0.21  | 0.36  | 0.49  | 0.47  | 0.43  | 0.27  | 0.33  | 0.31  | 0.26  | 0.17  | 0.08  | (45) |      |

## Historical Trends

|                              |       | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) Station Only            | DBAvg | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (47) Regional (38 neighbors) |       | (m)     |        |         |       |       |             |         |         |         |
|                              |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
|                              |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page